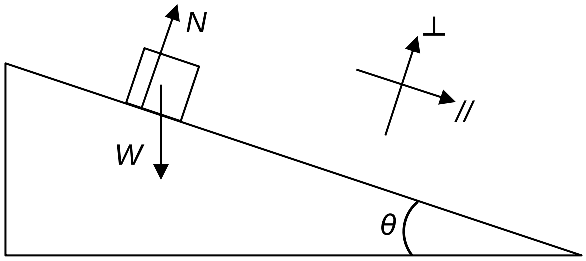
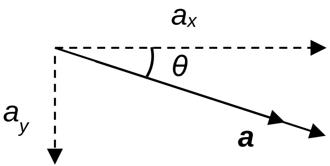
3. **Ans: C**





$$\overrightarrow{F_{net}} = m\vec{a}$$

$$\vec{W} + \vec{N} = m\vec{a}$$

$$\begin{pmatrix} W_{//} \\ W_{\perp} \end{pmatrix} + \begin{pmatrix} N_{//} \\ N_{\perp} \end{pmatrix} = m\begin{pmatrix} a_{//} \\ a_{\perp} \end{pmatrix}$$

$$\begin{pmatrix} mg\sin\theta \\ -mg\cos\theta \end{pmatrix} + \begin{pmatrix} 0 \\ N \end{pmatrix} = m\begin{pmatrix} a \\ 0 \end{pmatrix}$$

$$a = g\sin\theta$$

$$\begin{aligned} a_y &= a\sin\theta \\ &= (g\sin\theta)(\sin\theta) \\ &= g\sin^2\theta \end{aligned}$$